

1 From the electron–phonon interaction to the BCS ground state

1.1 Motivation

The chain of arguments leading from a lattice of ions and electrons to the BCS ground state is long, and no single source spells out every step: the derivation of the Fröhlich Hamiltonian, the canonical transformation that eliminates the phonons, the Cooper problem, the mean-field decoupling, the Bogoliubov diagonalization, and finally the pairing wavefunction. This note gives the skeleton of the argument together with the intermediate steps that are usually left as exercises: the choice of the generator S and its matrix elements, the sign structure that makes the induced interaction attractive, the evaluation of the Cooper and gap integrals, and the explicit verification that the BCS state is the quasiparticle vacuum.

The presentation follows G. Czycholl, *Theoretische Festkörperphysik* Band 2, Ch. 5; equation numbers of the form (C 5.x) refer to that book, so that the note can be read alongside it. The starting Hamiltonian is derived in Band 1, Sec. 7.1, and in Bruus & Flensberg, Ch. 3.

Two of the steps below are similarity transformations in the sense of Sec. ??, and are cross-referenced where they occur: the phonon elimination is a case where the Hadamard series does *not* close and must be truncated, whereas the Bogoliubov transformation is a textbook case where it does.

1.2 Physical picture

An electron e_1 moving through the lattice attracts the positively charged ions and leaves behind a region of enhanced positive charge density, i.e. a lattice distortion. Because the ions are slow ($\omega_D \ll E_F/\hbar$), this distortion persists after e_1 has moved on, and a second electron e_2 arriving later is attracted to it. The net result is an effective electron–electron interaction, mediated by the exchange of a phonon, that is attractive for electrons whose energies differ by less than a phonon energy. The retardation is essential: it is what allows the attraction to beat the (instantaneous, screened) Coulomb repulsion.

1.3 Step 1: the Fröhlich Hamiltonian

The starting point is

$$\begin{aligned}
 H = & \underbrace{\sum_{\vec{k},\sigma} \varepsilon_{\vec{k}} c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} + \sum_{\vec{q}} \hbar\omega_{\vec{q}} b_{\vec{q}}^\dagger b_{\vec{q}}}_{H_0} \\
 & + \underbrace{\sum_{\vec{k},\vec{q},\sigma} \left[M(\vec{q}) c_{\vec{k}+\vec{q},\sigma}^\dagger c_{\vec{k}\sigma} b_{\vec{q}} + M(-\vec{q}) c_{\vec{k}-\vec{q},\sigma}^\dagger c_{\vec{k}\sigma} b_{\vec{q}}^\dagger \right]}_{H_1}, \tag{1}
 \end{aligned}$$

with $\omega_{-\vec{q}} = \omega_{\vec{q}}$ and $M(-\vec{q}) = M(\vec{q})^*$ (the latter is forced by $H = H^\dagger$ after relabelling $\vec{k} \rightarrow \vec{k} - \vec{q}$ in the second term; for the jellium model M can be chosen real, which is the

convention used below). Each vertex absorbs or emits one phonon, conserves spin along the fermion line, and transfers momentum $\pm\vec{q}$.

1.4 Step 2: eliminating the phonons (Schrieffer–Wolff)

We want a purely electronic Hamiltonian, i.e. one that no longer contains processes changing the phonon number. Apply a canonical transformation

$$H_T = e^{-iS} H e^{iS}, \quad S^\dagger = S, \quad (2)$$

which is a similarity transformation with anti-Hermitian generator $A \equiv -iS$, so the Hadamard lemma of Sec. ?? applies:

$$H_T = H + [A, H] + \frac{1}{2}[A, [A, H]] + \dots \quad (3)$$

Choosing the generator. Split $H = H_0 + H_1$ and demand that the transformation cancel H_1 at first order,

$$H_1 + [A, H_0] = 0. \quad (4)$$

Everything at first order in the coupling is then gone, and collecting the remaining terms,

$$H_T = H_0 + \frac{1}{2}[A, H_1] + \mathcal{O}(H_1^3), \quad (5)$$

where the factor $\frac{1}{2}$ arises because $\frac{1}{2}[A, [A, H_0]] = -\frac{1}{2}[A, H_1]$ partially cancels $[A, H_1]$. Note the structure: *because* H_1 has been removed at first order, the leading effect is second order in M , which is exactly the phonon-exchange process.

Equation (4) is solved in the eigenbasis of H_0 : taking matrix elements of $[A, H_0]_{mn} = A_{mn}(E_n - E_m)$ gives

$$A_{mn} = \frac{(H_1)_{mn}}{E_m - E_n}. \quad (6)$$

Every term of H_1 in (1) is an eigenoperator of ad_{H_0} , so the energy denominators can be read off directly:

$$A = \sum_{\vec{k}, \vec{q}, \sigma} \left[\frac{M(\vec{q}) c_{\vec{k}+\vec{q}, \sigma}^\dagger c_{\vec{k}, \sigma} b_{\vec{q}}}{\varepsilon_{\vec{k}+\vec{q}} - \varepsilon_{\vec{k}} - \hbar\omega_{\vec{q}}} + \frac{M(\vec{q})^* c_{\vec{k}-\vec{q}, \sigma}^\dagger c_{\vec{k}, \sigma} b_{\vec{q}}^\dagger}{\varepsilon_{\vec{k}-\vec{q}} - \varepsilon_{\vec{k}} + \hbar\omega_{\vec{q}}} \right]. \quad (7)$$

The denominators are the energy cost of the corresponding virtual process. They are the origin of the resonant denominator in the final interaction, and of the fact that the transformation is ill-defined on resonance.

The effective interaction. Inserting A into (5), normal-ordering, and restricting to the zero-phonon subspace (this is the step where one “simply forbids” states with different phonon configurations from mixing) gives

$$H_{\text{int}} = \sum_{\vec{k}, \vec{k}', \vec{q}} \sum_{\sigma \sigma'} \frac{|M(\vec{q})|^2 \hbar\omega_{\vec{q}}}{(\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}+\vec{q}})^2 - (\hbar\omega_{\vec{q}})^2} c_{\vec{k}+\vec{q}, \sigma}^\dagger c_{\vec{k}', -\vec{q}, \sigma'}^\dagger c_{\vec{k}', \sigma'} c_{\vec{k}, \sigma}, \quad (8)$$

which is Czycholl's (C 5.25) written as

$$H_{\text{int}} = -\frac{1}{2} \sum_{\sigma\sigma'} \sum_{\vec{k}, \vec{k}', \vec{q}} V_{\vec{k}, \vec{k}', \vec{q}} c_{\vec{k}+\vec{q}, \sigma}^\dagger c_{\vec{k}', \sigma'}^\dagger c_{\vec{k}', \sigma'} c_{\vec{k}, \sigma}, \quad V_{\vec{k}, \vec{k}', \vec{q}} = \frac{2|M(\vec{q})|^2 \hbar \omega_{\vec{q}}}{(\hbar \omega_{\vec{q}})^2 - (\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}+\vec{q}})^2}. \quad (9)$$

The whole of superconductivity hangs on the sign of the denominator:

$$V_{\vec{k}, \vec{k}', \vec{q}} > 0 \iff |\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}+\vec{q}}| < \hbar \omega_{\vec{q}} \leq \hbar \omega_D, \quad (10)$$

i.e. the interaction is attractive exactly for electrons within a Debye energy of one another — a thin shell around the Fermi surface — and repulsive otherwise. The Debye frequency enters as the maximum available phonon energy.

Reading the vertex. In the diagram for (8), the wiggly line carries momentum $\vec{q}$ and each fermion line conserves its spin. Hence the incoming line $(\vec{k}, \sigma)$ must leave as $(\vec{k} + \vec{q}, \sigma)$ and $(\vec{k}', \sigma')$ as $(\vec{k}' - \vec{q}, \sigma')$. Connecting $(\vec{k}, \sigma)$ to $(\vec{k}' - \vec{q}, \sigma')$ is *not* a valid fermion line for this term: it would require a momentum transfer $\vec{k}' - \vec{q} - \vec{k} \neq \vec{q}$ in general, and a spin flip. That “crossed” assignment is the exchange process, which is a different term of the same sum (obtained by relabelling $\vec{q} \rightarrow \vec{k}' - \vec{q} - \vec{k}$) and is therefore already included — drawing it as a second contraction of the same term would double count.

Caveat. This elimination is *not* exact. H_1 is cubic in fermion/boson operators, so ad_A does not close on a finite set of operators (cf. the scope discussion in Sec. ??); the Hadamard series can only be truncated at $\mathcal{O}(M^2)$. Discarding the frequency dependence of the exchanged phonon in addition replaces a retarded interaction by an instantaneous one, which is what makes the naive comparison with the Coulomb repulsion misleading.

1.5 Step 3: the Cooper problem

Does an arbitrarily weak attraction destabilize the Fermi sea? Cooper's model: take a filled, inert Fermi sea

$$|\phi_0\rangle = \prod_{\varepsilon_{\vec{k}} \leq \varepsilon_F, \sigma} c_{\vec{k}\sigma}^\dagger |0\rangle \quad (11)$$

and add two electrons just above it, interacting only with each other via a constant attraction $V > 0$ in the shell $E_F \leq \varepsilon_{\vec{k}} \leq E_F + \hbar \omega_D$:

$$H_{\text{int}} = -\frac{V}{2} \sum_{\sigma\sigma'} \sum_{\vec{k}, \vec{q}} c_{\vec{k}+\vec{q}, \sigma}^\dagger c_{-\vec{k}-\vec{q}, \sigma'}^\dagger c_{-\vec{k}\sigma'} c_{\vec{k}\sigma}. \quad (12)$$

Only pairs with opposite momenta and opposite spins are retained (zero total momentum maximizes the phase space of scattering partners; the singlet is favoured because the orbital part is then symmetric and the two electrons can sit on top of each other, where the attraction acts). Make the variational ansatz

$$|\psi_{\sigma\sigma'}\rangle = \sum_{\vec{k}} a_{\sigma\sigma'}(\vec{k}) c_{\vec{k}\sigma}^\dagger c_{-\vec{k}\sigma'}^\dagger |\phi_0\rangle, \quad (13)$$

and minimize $E = \langle \psi | H | \psi \rangle$ subject to $\langle \psi | \psi \rangle \stackrel{!}{=} 1$ — this yields an *upper bound* on the energy of the bound state, which is all that is needed to prove instability.

Solving it. The stationarity condition is the two-body Schrödinger equation restricted to the shell,

$$2\varepsilon_{\vec{k}} a(\vec{k}) - V \sum_{\vec{k}'} a(\vec{k}') = E a(\vec{k}) \implies a(\vec{k}) = \frac{VC}{2\varepsilon_{\vec{k}} - E}, \quad C \equiv \sum_{\vec{k}'} a(\vec{k}'). \quad (14)$$

Summing over $\vec{k}$ divides out C and leaves the self-consistency condition

$$1 = V \sum_{\vec{k}} \frac{1}{2\varepsilon_{\vec{k}} - E} \simeq V g_0 \int_{E_F}^{E_F + \hbar\omega_D} \frac{d\varepsilon}{2\varepsilon - E} = \frac{V g_0}{2} \ln \frac{2E_F + 2\hbar\omega_D - E}{2E_F - E}, \quad (15)$$

where g_0 is the density of states *per spin* at E_F , taken constant across the thin shell. Writing $E = 2E_F - \Delta_b$ and solving for the binding energy,

$$\Delta_b = \frac{2\hbar\omega_D}{e^{2/(Vg_0)} - 1} \xrightarrow{Vg_0 \ll 1} 2\hbar\omega_D e^{-2/(Vg_0)}. \quad (16)$$

So (C 5.35): a bound state exists for *arbitrarily small* $V > 0$, lying below the non-interacting two-electron energy $2E_F$. The result is non-analytic in V — no order of perturbation theory in V would ever produce it. Since the argument applies to every pair near the Fermi edge, the normal Fermi sea is unstable, and one must treat the pairing self-consistently for all electrons at once. That is BCS theory.

1.6 Step 4: mean-field (BCS) Hamiltonian

Simplify (8) by keeping only opposite momenta and opposite spins and dropping the $\vec{q}$ -dependence of V (C 5.38):

$$H = \sum_{\vec{k}\sigma} \varepsilon_{\vec{k}} c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} - V \sum_{\vec{k}, \vec{k}'} c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger c_{-\vec{k}'\downarrow} c_{\vec{k}'\uparrow}, \quad |\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}'}| \leq \hbar\omega_D. \quad (17)$$

Now decouple the quartic term in the pair channel, $AB \approx \langle A \rangle B + A \langle B \rangle - \langle A \rangle \langle B \rangle$, with the pair amplitude

$$\Delta = V \sum_{\vec{k}} \langle c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow} \rangle. \quad (18)$$

Using $c_{-\vec{k}\downarrow}^\dagger c_{-\vec{k}\downarrow} = \mathbb{1} - c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow}^\dagger$ to bring the kinetic term into Nambu form gives (C 5.41)

$$H_{\text{MF}} = \sum_{\vec{k}} \begin{pmatrix} c_{\vec{k}\uparrow}^\dagger & c_{-\vec{k}\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_{\vec{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\vec{k}} \end{pmatrix} \begin{pmatrix} c_{\vec{k}\uparrow} \\ c_{-\vec{k}\downarrow}^\dagger \end{pmatrix} + \sum_{\vec{k}} \varepsilon_{\vec{k}} + \frac{|\Delta|^2}{V}. \quad (19)$$

Here $\varepsilon_{\vec{k}}$ must be measured from the chemical potential, $\varepsilon_{\vec{k}} \rightarrow \varepsilon_{\vec{k}} - \mu$, since H_{MF} does not conserve particle number and one is really working in the grand canonical ensemble.

1.7 Step 5: Bogoliubov diagonalization

Introduce a unitary $U_{\vec{k}}$ diagonalizing the 2×2 matrix, so that H_{MF} contains only number operators of new fermions. Taking Δ real and positive (its phase can be absorbed into the fermion operators) the eigenvectors, normalized to one, give (C 5.43)

$$\alpha_{\vec{k}} = u_{\vec{k}} c_{\vec{k}\uparrow} - v_{\vec{k}} c_{-\vec{k}\downarrow}^\dagger, \quad \beta_{\vec{k}} = u_{\vec{k}} c_{-\vec{k}\downarrow} + v_{\vec{k}} c_{\vec{k}\uparrow}^\dagger, \quad (20)$$

$$u_{\vec{k}}^2 = \frac{1}{2} \left(1 + \frac{\varepsilon_{\vec{k}}}{E_{\vec{k}}} \right), \quad v_{\vec{k}}^2 = \frac{1}{2} \left(1 - \frac{\varepsilon_{\vec{k}}}{E_{\vec{k}}} \right), \quad 2u_{\vec{k}}v_{\vec{k}} = \frac{\Delta}{E_{\vec{k}}}, \quad u_{\vec{k}}^2 + v_{\vec{k}}^2 = 1. \quad (21)$$

Because $U_{\vec{k}}$ is unitary, the new operators are again fermionic, $\{\alpha_{\vec{k}}, \alpha_{\vec{k}'}^\dagger\} = \delta_{\vec{k}\vec{k}'}$, and hence $\langle \alpha_{\vec{k}}^\dagger \alpha_{\vec{k}} \rangle = f(E_{\vec{k}})$ (Fermi function) while anomalous averages such as $\langle \beta_{\vec{k}} \alpha_{\vec{k}} \rangle$ vanish. The result is (C 5.46)–(C 5.47)

$$H_{\text{eff}} = \sum_{\vec{k}} E_{\vec{k}} (\alpha_{\vec{k}}^\dagger \alpha_{\vec{k}} + \beta_{\vec{k}}^\dagger \beta_{\vec{k}}) + \text{const}, \quad E_{\vec{k}} = (\varepsilon_{\vec{k}}^2 + |\Delta|^2)^{1/2} \geq 0, \quad (22)$$

with $\text{const} = \sum_{\vec{k}} (\varepsilon_{\vec{k}} - E_{\vec{k}}) + |\Delta|^2/V$. The excitation spectrum is gapped: $E_{\vec{k}} \geq |\Delta|$ for all $\vec{k}$.

As a similarity transformation. The same result follows from Method B of Sec. ?? . Take

$$S = \sum_{\vec{k}} \theta_{\vec{k}} (c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger - c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow}), \quad S^\dagger = -S. \quad (23)$$

Then ad_S closes on the pair $\{c_{\vec{k}\uparrow}, c_{-\vec{k}\downarrow}^\dagger\}$,

$$[S, c_{\vec{k}\uparrow}] = -\theta_{\vec{k}} c_{-\vec{k}\downarrow}^\dagger, \quad [S, c_{-\vec{k}\downarrow}^\dagger] = +\theta_{\vec{k}} c_{\vec{k}\uparrow}, \quad (24)$$

so $M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is elliptic and the flow is a rotation,

$$e^S c_{\vec{k}\uparrow} e^{-S} = \cos \theta_{\vec{k}} c_{\vec{k}\uparrow} - \sin \theta_{\vec{k}} c_{-\vec{k}\downarrow}^\dagger, \quad (25)$$

i.e. $u_{\vec{k}} = \cos \theta_{\vec{k}}$, $v_{\vec{k}} = \sin \theta_{\vec{k}}$ with $\tan 2\theta_{\vec{k}} = \Delta/\varepsilon_{\vec{k}}$, reproducing (21). This is the fermionic counterpart of the (hyperbolic) bosonic squeezing transformation; the compact rotation instead of a boost is the signature of the anticommuting algebra.

1.8 Step 6: the gap equation

Feed the diagonalization back into the definition (18). Inverting, $c_{\vec{k}\uparrow} = u_{\vec{k}} \alpha_{\vec{k}} + v_{\vec{k}} \beta_{\vec{k}}^\dagger$ and $c_{-\vec{k}\downarrow} = u_{\vec{k}} \beta_{\vec{k}} - v_{\vec{k}} \alpha_{\vec{k}}^\dagger$, so

$$\langle c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow} \rangle = u_{\vec{k}} v_{\vec{k}} (1 - 2f(E_{\vec{k}})) = \frac{\Delta}{2E_{\vec{k}}} \tanh\left(\frac{\beta E_{\vec{k}}}{2}\right), \quad (26)$$

($\beta = 1/k_B T$ here — note the unfortunate clash with the Bogoliubov operator $\beta_{\vec{k}}$), giving the BCS self-consistency equation (C 5.50)

$$\Delta = \frac{V\Delta}{2} \sum_{\vec{k}} \frac{1}{E_{\vec{k}}} \tanh\left(\frac{\beta E_{\vec{k}}}{2}\right). \quad (27)$$

$\Delta = 0$ always solves this. For $\Delta \neq 0$ divide through (C 5.51):

$$1 = \frac{V}{2} \sum_{\vec{k}}^{\|\varepsilon_{\vec{k}}\| < \hbar\omega_D} (\varepsilon_{\vec{k}}^2 + |\Delta|^2)^{-1/2} \tanh\left(\frac{\beta(\varepsilon_{\vec{k}}^2 + |\Delta|^2)^{1/2}}{2}\right). \quad (28)$$

Two limits worth having explicitly. At $T = 0$ the $\tanh \rightarrow 1$ and, with g_0 constant,

$$1 = \frac{Vg_0}{2} \int_{-\hbar\omega_D}^{\hbar\omega_D} \frac{d\varepsilon}{\sqrt{\varepsilon^2 + \Delta^2}} = Vg_0 \operatorname{arcsinh} \frac{\hbar\omega_D}{\Delta} \implies \Delta(0) = \frac{\hbar\omega_D}{\sinh(1/(Vg_0))} \simeq 2\hbar\omega_D e^{-1/(Vg_0)}. \quad (29)$$

Setting instead $\Delta \rightarrow 0$ locates the critical temperature, $k_B T_c \simeq 1.13 \hbar\omega_D e^{-1/(Vg_0)}$, whence the parameter-free BCS ratio $2\Delta(0)/k_B T_c \simeq 3.53$.

Comparing with the Cooper result (16): the exponents differ by a factor of two, $e^{-1/(Vg_0)}$ versus $e^{-2/(Vg_0)}$. The Cooper binding energy of a single pair on top of a rigid Fermi sea is therefore *not* the gap; the self-consistent treatment, in which all pairs form simultaneously and the sea rearranges, is parametrically more strongly bound.

Why Δ is an order parameter. In a normal metal $\langle c_{\vec{k}\uparrow} c_{-\vec{k}\downarrow} \rangle = 0$ for all $\vec{k}$, because a state of sharp particle number gives zero expectation to an operator that changes N by two; hence $\Delta = 0$. A solution with $\Delta \neq 0$ therefore signals a state that is not an eigenstate of N , i.e. one with a condensate of Cooper pairs and a spontaneously chosen phase. Δ is complex, and its phase is the broken $U(1)$ symmetry.

1.9 Step 7: the BCS ground state

Since $E_{\vec{k}} \geq 0$ and $\alpha_{\vec{k}}^\dagger \alpha_{\vec{k}}$, $\beta_{\vec{k}}^\dagger \beta_{\vec{k}}$ have eigenvalues 0, 1, the ground state of H_{eff} is the state containing no quasiparticles:

$$\alpha_{\vec{k}} |\text{BCS}\rangle = \beta_{\vec{k}} |\text{BCS}\rangle = 0 \quad \forall \vec{k}. \quad (30)$$

This is satisfied by

$$|\text{BCS}\rangle = \prod_{\vec{k}} (u_{\vec{k}} + v_{\vec{k}} c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger) |0\rangle = \bigotimes_{\vec{k}} (u_{\vec{k}} + v_{\vec{k}} c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger) |0\rangle_{\vec{k}}. \quad (31)$$

Verification. Two facts make this a one-line check. First, each factor $P_{\vec{k}} \equiv u_{\vec{k}} + v_{\vec{k}} c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger$ is *even* in fermion operators, hence commutes with everything; so $\alpha_{\vec{k}}$ can be moved past all

factors $P_{\vec{k}'}$, $\vec{k}' \neq \vec{k}$, without sign bookkeeping and acts on $P_{\vec{k}}|0\rangle_{\vec{k}}$ alone. Second, within that single mode,

$$u_{\vec{k}}c_{\vec{k}\uparrow}P_{\vec{k}}|0\rangle = u_{\vec{k}}v_{\vec{k}}c_{\vec{k}\uparrow}c_{\vec{k}\uparrow}^{\dagger}c_{-\vec{k}\downarrow}^{\dagger}|0\rangle = u_{\vec{k}}v_{\vec{k}}c_{-\vec{k}\downarrow}^{\dagger}|0\rangle, \quad (32)$$

$$-v_{\vec{k}}c_{-\vec{k}\downarrow}^{\dagger}P_{\vec{k}}|0\rangle = -u_{\vec{k}}v_{\vec{k}}c_{-\vec{k}\downarrow}^{\dagger}|0\rangle \quad (\text{the } v^2 \text{ term dies on } (c_{-\vec{k}\downarrow}^{\dagger})^2 = 0), \quad (33)$$

and the two cancel, so $\alpha_{\vec{k}}|\text{BCS}\rangle = 0$. The computation for $\beta_{\vec{k}}$ is identical up to one anti-commutation sign. The tensor-product form in (31) reflects that the Hilbert space factorizes into independent four-dimensional blocks $\{|0\rangle, c_{\vec{k}\uparrow}^{\dagger}|0\rangle, c_{-\vec{k}\downarrow}^{\dagger}|0\rangle, c_{\vec{k}\uparrow}^{\dagger}c_{-\vec{k}\downarrow}^{\dagger}|0\rangle\}$, of which the pairing state occupies only the empty and doubly occupied ones.

Properties. Normalization is automatic, $\langle \text{BCS} | \text{BCS} \rangle = \prod_{\vec{k}} (u_{\vec{k}}^2 + v_{\vec{k}}^2) = 1$, and

$$\langle n_{\vec{k}\uparrow} \rangle = v_{\vec{k}}^2 = \frac{1}{2} \left(1 - \frac{\varepsilon_{\vec{k}}}{E_{\vec{k}}} \right), \quad \langle c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow} \rangle = u_{\vec{k}} v_{\vec{k}} = \frac{\Delta}{2E_{\vec{k}}}. \quad (34)$$

The step function of the Fermi sea is smeared over a width $\sim \Delta$ about $\varepsilon_{\vec{k}} = 0$, and $u_{\vec{k}}v_{\vec{k}}$ (the coherence factor) is peaked exactly there: only the shell where occupation is uncertain contributes to pairing. Particle number is not sharp, $\langle N \rangle = 2 \sum_{\vec{k}} v_{\vec{k}}^2$ with $\langle \Delta N^2 \rangle = 4 \sum_{\vec{k}} u_{\vec{k}}^2 v_{\vec{k}}^2 = \mathcal{O}(N)$, so relative fluctuations are $\mathcal{O}(N^{-1/2})$ and harmless in the thermodynamic limit.

Remark: a pair-product wavefunction. Factoring out $\prod u_{\vec{k}}$ recasts (31) as a coherent state of pairs,

$$|\text{BCS}\rangle \propto \exp \left(\sum_{\vec{k}} g_{\vec{k}} c_{\vec{k}\uparrow}^{\dagger} c_{-\vec{k}\downarrow}^{\dagger} \right) |0\rangle, \quad g_{\vec{k}} = \frac{v_{\vec{k}}}{u_{\vec{k}}}, \quad (35)$$

whose fixed- N projection is the geminal (pair-product) wavefunction $(\sum_{\vec{k}} g_{\vec{k}} c_{\vec{k}\uparrow}^{\dagger} c_{-\vec{k}\downarrow}^{\dagger})^{N/2} |0\rangle$. This is the form in which BCS pairing is usually grafted onto variational many-body ansätze: a Pfaffian/geminal reference state, optionally multiplied by a correlator or a neural-network amplitude.

1.10 Pitfalls and conventions

- **Sign of the Cooper result.** With $\Delta_b \equiv 2E_F - E_{\text{bs}} > 0$ the binding energy is positive, $\Delta_b \simeq +2\hbar\omega_D e^{-2/(Vg_0)}$; sources that write $\Delta E \simeq -2\hbar\omega_D e^{-2/(Vg_0)}$ are defining $\Delta E = E_{\text{bs}} - 2E_F$. Worth fixing one convention, as the two appear interchangeably in the literature.
- **Density of states.** g_0 above is per spin. Using the total DOS ($2g_0$) silently halves the exponents.
- $\varepsilon_{\vec{k}}$ vs. $\varepsilon_{\vec{k}} - \mu$. From Step 4 onwards energies are measured from the chemical potential; the gap equation is wrong otherwise.
- **Overloaded β .** Inverse temperature and Bogoliubov quasiparticle share the symbol in the standard notation.

- **Validity.** The phonon elimination is second order in M and breaks down on resonance; the mean-field decoupling neglects pair fluctuations (fine in $d = 3$ with the long coherence length of conventional superconductors, not in general); and the Coulomb repulsion, omitted throughout, must be screened and retardation-reduced (Morel–Anderson μ^*) before the net interaction is attractive.

1.11 References

- G. Czycholl, *Theoretische Festkörperphysik* Band 1, Sec. 7.1 (Fröhlich Hamiltonian) and Band 2, Ch. 5 (Cooper pairs, BCS).
- H. Bruus & K. Flensberg, *Many-Body Quantum Theory in Condensed Matter Physics*, Ch. 3 and Ch. 18.
- J. Bardeen, L.N. Cooper & J.R. Schrieffer, Phys. Rev. **108**, 1175 (1957).
- L.N. Cooper, Phys. Rev. **104**, 1189 (1956).
- M. Tinkham, *Introduction to Superconductivity*, Ch. 3.
- A. Fetter & J.D. Walecka, *Quantum Theory of Many-Particle Systems*, Ch. 13.
- P. Coleman, *Introduction to Many-Body Physics*, Ch. 14–15.
- J.R. Schrieffer & P.A. Wolff, Phys. Rev. **149**, 491 (1966).